



INF212

ALGORITHMS AND PROGRAMMING

PROJECT-1

STUDENT PERFORMANCE EVALUATION SYSTEM

The deadline is Monday, March 23, 2026, at 23:59.

Projects that are not delivered on time are not accepted.

Upload the project to the Project 1 assignment section of the INF212 class.

The questions can be asked to course lecturer Dr. Murat Ceylan and teaching assistant Mehmet Emin GEÇER.

PROJECT 1

PROBLEM DEFINITION

The problem addressed in this project is the development of a simple, reliable, and efficient software system that can manage and analyze student performance data. The system must accept multiple exam inputs for each student, compute a weighted success score, classify students based on predefined criteria, and present meaningful statistical results.

The program should also allow users to interact with the system through a menu-driven interface, enabling different types of analysis such as listing student scores, sorting performance results, counting grade distributions, and calculating statistical measures. The system must ensure data validity by restricting inputs to acceptable ranges and must operate correctly for different numbers of students.

OBJECTIVE

The aim of this project is to develop a menu-driven program in C that analyzes students' exam scores.

This project is designed to improve students' skills in:

- array usage
- function design
- loop structures
- conditional statements
- basic statistical computations

PROGRAM REQUIREMENTS

The program must be menu-driven and include the following features:

1. List all student scores
2. Calculate class average
3. Find highest and lowest scores
4. Display the number of Pass / Make-up / Fail students
5. List letter grades of students
6. Sort students by success score (descending order)
7. Count students according to letter grades
8. Calculate standard deviation
9. Exit program

REQUIRED PROGRAMMING STRUCTURES

1. The number of students must be entered by the user.
2. Scores must be validated (between 0 and 100).
3. The program must run until the user selects exit.
4. Output must be clear and well-formatted.

The following concepts must be used:

- Student data must be stored using arrays: student IDs, midterm scores, final scores, quiz scores, success scores
- Use for, while, or do-while loops where necessary. Use if-else and switch-case structures.
- The program must be modular and divided into functions.

At minimum, the program should include functions for:

- inputting scores
- listing scores
- calculating average
- finding maximum score
- finding minimum score
- counting pass/make-up/fail students
- sorting students
- counting letter grades
- calculating standard deviation

(Important Note: The following are **NOT allowed** in this project: Pointers, Structures (struct), File operations. Students must only use: arrays, loops, functions, conditional statements)

SUCCESS SCORE CALCULATION

The overall success score is calculated as:

$$\text{Success Score} = 0.3 \times \text{Midterm} + 0.5 \times \text{Final} + 0.2 \times \text{Quiz}$$

STANDARD DEVIATION

The standard deviation of the class must be calculated using:

$$\sigma = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}$$

where:

- x_i : student score
- $\bar{x}$: class average
- n : number of students

PERFORMANCE CRITERIA

- **60 and above** → Pass
- **50 – 59** → Make-up Exam
- **Below 50** → Fail

LETTER GRADE SYSTEM

- **90 – 100** → AA
- **85 – 89** → BA
- **80 – 84** → BB
- **70 – 79** → CB
- **60 – 69** → CC
- **50 – 59** → DC
- **0 – 49** → FF